

Rohith Narsingipyta

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Education

Vasavi College of Engineering — *B.E. (Honours) in Electronics and Communication*

Expected 2027

Specialisation: System on Chip Design (GPA: 8.0/10)

- **Honours Coursework:** SystemVerilog, UVM, AXI Protocol, AMBA Protocols

Technical Skills

Hardware & EDA: Xilinx Vivado, Cadence Virtuoso, Synopsys VCS, Verdi, Zynq
Embedded & OS: Yocto, Buildroot, QNX, Linux
Languages: Verilog, SystemVerilog, C, C++, Go, Python
Frameworks & Tools: Qt, GStreamer, Pion, SWUpdate, Git, CMake, MongoDB

Experience

Aervia — *Founding Embedded Systems Engineer*

6 Months

Architected Production Headless Embedded AI Voice Agent (FRDM-i.MX93)

- Built a custom Yocto BSP layer for the FRDM-i.MX93, removing the graphics stack and unused packages to create a lightweight headless image with reduced boot time.
- Owned the end-to-end implementation of a native Go (Pion) WebRTC client; bypassed standard wrappers to establish low-overhead audio streaming connecting directly to the OpenAI Realtime API.
- Achieved architectural latency boundaries of sub-500ms (voice) and sub-1000ms (with custom backend tool calls) at a strict 20% memory utilization ceiling.
- Designed a fault-tolerant A/B OTA update infrastructure using SWUpdate coupled with a read-only rootfs to enforce secure, atomic full-image production updates.
- Integrated GStreamer audio pipelines with vendor-binary Bluetooth SCO routing; managed structured logging via Zap/Lumberjack and successfully deployed final firmware image to 5 pilot production devices.

Projects

3×3 Mesh Network-on-Chip (NoC)

Verilog, Vivado

- Defined the microarchitecture of a packet-switched 3×3 mesh NoC; implemented a custom 32-bit flit structure (8-bit payload), virtual channels, and round-robin arbitration logic.
- Engineered XY routing algorithms and verified end-to-end packet delivery using testbenches with comparative payload checks against expected data paths.

AXI DMA Controller on Zynq

Verilog, C, Zynq, Vivado

- Designed an AXI DMA controller utilizing AXI Stream and master/slave hardware blocks in Verilog; wrote memory-mapped (mmap) PS-side drivers and test code in C to manage data transfers.
- Verified pipeline data integrity across the hardware framework utilizing sequential and random data patterns to catch transaction constraints.

Embedded SIP Communication System on STM32

C++, Qt, Buildroot, SIP

- Constructed a multi-node communication topology on an embedded Linux (Buildroot) base using four STM32MP157F boards, anchoring one Asterisk server to route three client streams.
- Programmed interactive custom Qt GUIs to interface server-side real-time call telemetry and client-side touchscreen UI contact lists seamlessly.

Leadership

Silicon Synapses, Vasavi College of Engineering

Sub-Lead

- Conducted a hands-on FPGA workshop for 50+ students, teaching digital logic and FPGA programming from scratch with live build-along sessions
- Organized a VLSI design competition involving circuit debugging and FPGA implementation challenges